

SAKUM_HINGLISH.md — Canonical Hinglish Keyword Glossary (single source)

Per user doctrine (2026-07-18): Sakum Lang keywords are **Hinglish (romanized Sanskrit) only** — typeable ASCII, no Devanagari required, no pure-English keywords. This file is the ONE place the keyword spelling is defined. Every other doc, the assembly registry (`assembly/sakum_keywords.s`), and the compiler lexer must agree with this table.

1. Core language keywords

Hinglish keyword	Meaning	Replaces (old)
aarambh	program / main	आरम्भ
naam	declare variable	नाम / चर
kriya	function	क्रिया / सूत्र
yadi	if / conditional	यदि
anyatha	else	अन्यथा
yavat	loop (while)	यावत् / जबतक
paryantam	counted loop (for)	पर्यन्तम्
vapsa	return	प्रत्यागम / वापस
satya	true	सत्य
asatya	false	असत्य
shunya	nil / none	शून्य
lek	output / print	लेख / मुद्रण
hriday	engine allocator	हृदय
spand	engine tick	स्पन्द
naadi	signal bus	नाडी
sutra	creator key / thread	सूत्र / सूत्र
varg	class	वर्ग
vyakhya	explain error	व्याख्या
svadhyaya	self-learn	स्वाध्याय

Hinglish keyword	Meaning	Replaces (old)
santripati	saturating arith	संतृप्ति:
vektor	SIMD vector	वेक्टर
ank	integer elem type	अङ्क
paripath	quantum circuit	परिपथ
dvar	quantum gate	द्वार
map	measure	माप
brahma	web crawler	ब्रम्ह
pariksha	self-test block	परीक्षा
aur	logical and	और
athava	logical or	अथवा
lambai	length	लंबाई
sanchay	database engine	सञ्चय
kech	key/value store	केच
anukra	vectorless index	अनुक्र
grantha	graph store	ग्रन्थ
tantra	systems engineering	तन्त्र
anvesh	search / lookup	अन्वेष
sarani	hash table	सारणी
chakra	ring buffer	चक्र
pravah	pipe operator	>

2. Domain keyword registry (148 reserved, Hinglish spellings)

The 148 domain keywords keep their romanized (Hinglish) spellings already used in `assembly/sakum_keywords.s` (e.g. `vastu`, `sutra`, `hriday`, `prajna`).

They are reserved on every target. Category ids (0..12) are unchanged:

```
0 TYPES  1 FUNC  2 CONC  3 MEM   4 FS    5 NET   6 AI
7 ROBOT  8 QUANT 9 COMPILER 10 SEC 11 DIST 12 LIVING
```

The full 148-row table lives in `assembly/sakum_keywords.s` (machine-level registry) and is mirrored in `spec/spec_keywords.sak`.

3. New machine-level DB shapes (requirement 2 & 3)

The `sanchay` engine implements FIVE primitive data shapes, all at machine level in `assembly/sakum_db.s` :

Hinglish	Shape	Notes
kech	key/value	sutra-encrypted at rest
vektor	vector (ANN)	SIMD L2 distance, stateful index
anukra	vectorless	B-tree / inverted classical index
sthit	stateful store	mutable, persisted, addressable by hash
asthit	stateless store	pure key->value, no persistence
grantha	graph	typed edges + naadi-driven traversal

Note: `sthit` (stateful) and `asthit` (stateless) are NEW shapes added to the `sanchay` engine per the user's requirement. They join the original `kech` / `vektor` / `anukra` / `grantha` four.